

1. 二进制 GA 求解函数优化问题 MATLAB 代码

func1

```
function result=func1(x)
fit= x+10*sin(5*x)+7*cos(4*x);
result=fit;
```

GA

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%标准遗传算法求函数极值%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%初始化参数%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
clear all;                %清除所有变量
close all;                %清图
clc;                      %清屏
NP=50;                    %种群数量
L=20;                     %二进制数串长度
Pc=0.8;                   %交叉率
Pm=0.1;                   %变异率
G=100;                    %最大遗传代数
Xs=10;                    %上限
Xx=0;                     %下限
f=randi([0,1],NP,L);      %随机获得初始种群
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%遗传算法循环%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
for k=1:G
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%将二进制解码为定义域范围内十进制%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    for i=1:NP
        U=f(i,:);
        m=0;
        for j=1:L
            m=U(j)*2^(j-1)+m;
        end
        x(i)=Xx+m*(Xs-Xx)/(2^L-1);
        Fit(i)= func1(x(i));
    end
    maxFit=max(Fit);        %最大值
    minFit=min(Fit);        %最小值
    rr=find(Fit==maxFit);
    fBest=f(rr(1,1),:);     %历代最优个体
    xBest=x(rr(1,1));
    Fit=(Fit-minFit)/(maxFit-minFit); %归一化适应度值
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%基于轮盘赌的复制操作%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    sum_Fit=sum(Fit);
    fitvalue=Fit./sum_Fit;
    fitvalue=cumsum(fitvalue);
    ms=sort(rand(NP,1));
    fiti=1;
```

```

newi=1;
while newi<=NP
    if (ms(newi))<fitvalue(fiti)
        nf(newi,:)=f(fiti,:);
        newi=newi+1;
    else
        fiti=fiti+1;
    end
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%基于概率的交叉操作%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
for i=1:2:NP
    p=rand;
    if p<Pc
        q=randi([0,1],1,L);
        for j=1:L
            if q(j)==1;
                temp=nf(i+1,j);
                nf(i+1,j)=nf(i,j);
                nf(i,j)=temp;
            end
        end
    end
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%基于概率的变异操作%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
i=1;
while i<=round(NP*Pm)
    h=randi([1,NP],1,1); %随机选取一个需要变异的染色体
    for j=1:round(L*Pm)
        g=randi([1,L],1,1); %随机需要变异的基因数
        nf(h,g)=~nf(h,g);
    end
    i=i+1;
end
f=nf;
f(1,:)=fBest; %保留最优个体在新种群中
trace(k)=maxFit; %历代最优适应度
end
xBest; %最优个体
figure
plot(trace)
xlabel('迭代次数')
ylabel('目标函数值')
title('适应度进化曲线')

```